

YITIAN WANG

+1(929)624-1021
ywang00149@gmail.com

cosmotim.github.io
[LinkedIn](#)

SUMMARY

Thermal-sciences researcher with a Ph.D. in Electrical and Computer Engineering and expertise in heat transfer, thermodynamics, thermal transport, and electrochemistry for solid-state energy materials. Experienced in designing and executing experimental programs, developing instrumentation and data-reduction workflows, and applying analytical and numerical modeling using Python, MATLAB, and COMSOL. Published 6 first-author and 3 co-authored peer-reviewed journal articles; experienced in national-laboratory collaboration, technical proposals, and communicating results to cross-disciplinary research teams.

EDUCATION

University of California, Riverside, CA Ph.D. in Electrical and Computer Engineering	Mar. 2021 – Jun. 2026
Columbia University in the City of New York, NY Master of Science, Material Science and Engineering	Aug. 2019 – Feb. 2021
University of California, Berkeley, CA Summer Session, Physics	Jul. 2017 – Aug. 2017
Beijing Normal University, CN Bachelor of Science, Physics	Sep. 2015 – Jun. 2019

SKILLS

Thermal Sciences & Modeling	Heat Transfer, Thermodynamics, Thermal Transport, Python, MATLAB, COMSOL, SolidWorks
Software	R, SQL, LaTeX, Linux
Analysis Tools	GSAS-II, Mantid, ImageJ, VESTA, Quantum ESPRESSO, VASP
Synthesis & Growth	Floating-Zone Crystal Growth, PVD / Magnetron Sputtering, CVD
Characterization	SEM, TEM, Scanning Tunneling Microscopy (STM), XRD, AFM, Inelastic Neutron Scattering, Raman, FTIR
Transport Properties	PPMS, DSC, TGA, Four-Probe Transport, Mechanical Strength Testing
Battery & Electrochem	Prototype Cell Assembly, Glovebox Operation, Separator Fabrication, EIS, Galvanostatic Cycling, Thermal Abuse Testing
Methods	Design of Experiments (DOE), AI Agent Training

AWARDS & ACTIVITIES

- Dissertation Completion Fellowship Award, University of California, Riverside (2025)
- MRS 2023 Spring Meeting Highly Commended Student Talk Award
- ‘Jingshi’ Scholarship, Beijing Normal University (2016 – 2018)
- COMAP Mathematical Contest in Modeling — team member and team leader

PROFESSIONAL MEMBERSHIPS

- Sigma Xi, The Scientific Research Honor Society — Elected Full Member (2026)
- American Physical Society (APS) — Member (2025 – 2026)

- Materials Research Society (MRS) — Student Member (through 2027)

RESEARCH EXPERIENCE

Postdoctoral Fellow

The University of Texas at Austin

Jul. 2026 – Present

Austin, TX

- Conducting postdoctoral research in materials and energy-related engineering.

Graduate Research Assistant

University of California

Jun. 2021 – Jun. 2025

Riverside, CA

- Designed and executed thermal-transport experiments on Li-ion solid-state electrolytes, extending lattice-dynamics studies previously limited to Cu- and Ag-ion systems.
- Published 6 first-author papers in journals including *Applied Physics Reviews*, *PRX Energy*, and *J. Mater. Chem. A*.
- Developed a floating-zone crystal-growth workflow and experimental specimens, producing centimeter-scale LLZTO single crystals for phonon-resolved neutron-scattering measurements.
- Identified *diffuson*-mediated thermal transport in Li-ion conductors, providing the first verification of this mechanism in Li-ion solid-state electrolytes.
- Developed a two-channel (phonon + *diffuson*) thermal-conductivity model in MATLAB that explained anomalous temperature dependence in Li-ion conductors where the traditional Debye model failed.

Experiment Lead

Oak Ridge National Laboratory

Jan. 2022 – Jul. 2023

Oak Ridge, TN

- Authored two successful beam time proposals and led inelastic neutron-scattering experiments on LLZTO single crystals using TAX (HFIR) and ARCS (SNS) at Oak Ridge National Laboratory (ORNL).
- Resolved the full phonon dispersion of a garnet solid electrolyte for the first time, providing experimental benchmarks for ab initio lattice-dynamics calculations.
- Built automated data-acquisition reduction and analysis pipelines in Mantid and Python, improving reproducibility and reducing manual analysis time.
- Led on-site experimental decisions and coordinated a multi-institutional team spanning ~7 research groups at UCR, UT Austin, ORNL, ETH Zürich, and TU Wien.

Student Researcher

Columbia University

Sep. 2019 – Jan. 2021

New York, NY

- Engineered composite Li-ion battery separators (γ -C₃N₄/PVDF) via solution processing, targeting improved electrochemical reaction kinetics.
- Assembled and tested CR2032 full cells in an Ar-filled glovebox over 100+ galvanostatic cycles, characterizing cycling performance and interfacial stability via EIS.
- Applied first-principles DFT (Quantum ESPRESSO) to model defect stability in olivine cathode structures, complementing experimental work with computational materials design.

Student Researcher

Beijing Normal University

Sep. 2018 – May 2019

Beijing, CN

- Designed a uniaxial-strain apparatus in SolidWorks exploiting differential thermal contraction for controlled detwinning of iron-based superconductors below 140 K.
- Built a Monte Carlo simulation in MATLAB to model atomic movement of metal atoms inside multi-wall carbon nanotubes, reproducing experimental observations.

TEACHING EXPERIENCE

Graduate Teaching Assistant

University of California, Riverside

Technology in the Premodern World (ENGR 108)

Reader

Fall 2025

N/A

Circuits and Electronics Lab (EE 005)

Lab Instructor

Spring 2025

7.0/7.0 (96th %ile)

Technology in the Premodern World (ENGR 108)

Teaching Assistant

Spring 2025

5.7/7.0 (83rd %ile)

Linear Methods for Engr. Analysis – MATLAB (EE 020B)

Teaching Assistant

Winter 2023

6.5/7.0 (81st %ile)

Magnetic Materials (EE 139)

Teaching Assistant

Fall 2022

6.5/7.0 (77th %ile)

- Served 280+ students across 7 sections over 3 academic years.
- Achieved perfect 7.0/7.0 evaluation (96th percentile campus-wide) in EE 005, with consistently above-department averages across all courses.
- Selected student feedback: “singlehandedly one of the most effective teachers on this campus” and “top tier educator” (EE 005); “the backbone of the entire class” (EE 020B); “always helpful... feedback was helpful in creating a well organized presentation” (EE 139).

INTERNSHIP EXPERIENCE

Administrative Assistant, Physics Department

Beijing Normal University

Sep. 2018 – Jan. 2019

Beijing, CN

- Managed archival of 3 years of departmental paperwork and organized the academic calendar for 4 course sections.

Student Intern

Shanghai East China Computer Corp.

Jul. 2016 – Aug. 2016

Shanghai, CN

- Gained hands-on exposure to IT industry workflows, strengthening teamwork and cross-functional communication skills.

SELECTED PUBLICATIONS

1. **Y. Wang**, H. Wang, X. Chen. “Thermal transport mechanisms in electrolytes for all-solid-state batteries.” *Appl. Phys. Rev.*, Sep. 2026. [\[Link\]](#)
2. **Y. Wang**, Q. Jia, S. Li, L. Shi, Y. Li, X. Chen. “Low thermal conductivity and lattice anharmonicity of NaSICON-type solid electrolyte $\text{Na}_3\text{Zr}_2\text{Si}_2\text{PO}_{12}$.” *Tungsten*, Nov. 2025. [\[Link\]](#)
3. **Y. Wang**, Q. Jia, S. Li, L. Shi, Y. Li, X. Chen. “Glass-like thermal transport in polycrystalline perovskite Li-ion conductor $\text{Li}_{3/8}\text{Sr}_{7/16}\text{Hf}_{1/4}\text{Ta}_{3/4}\text{O}_3$.” *Chem. Commun.*, Sep. 2025. [\[Link\]](#) [\[Front Cover\]](#)
4. **Y. Wang**, Y. Su, J. Carrete, H. Zhang, N. Wu, *et al.* “Origin of intrinsically low thermal conductivity in a garnet-type solid electrolyte.” *PRX Energy*, Jul. 2025 (OA fee waived). [\[Link\]](#)
5. S. Li, S. Guo, **Y. Wang**, H. Li, Y. Xu, V. Carta, J. Zhou, X. Chen. “Enhanced magnon thermal transport in yttrium-doped spin ladder compounds $\text{Sr}_{14-x}\text{Y}_x\text{Cu}_{24}\text{O}_{41}$.” *J. Appl. Phys.*, Jul. 2024. [\[Link\]](#)
6. **Y. Wang**, S. Li, N. Wu, Q. Jia, T. Hoke, L. Shi, Y. Li, X. Chen. “Thermal properties and lattice anharmonicity of Li-ion conducting garnet solid electrolyte $\text{Li}_{6.5}\text{La}_3\text{Zr}_{1.5}\text{Ta}_{0.5}\text{O}_{12}$.” *J. Mater. Chem. A*, Jun. 2024. [\[Link\]](#)
7. S. Guo, H. Li, X. Bai, **Y. Wang**, S. Li, R. E. Dunin-Borkowski, J. Zhou, X. Chen. “Size-dependent magnon thermal transport in a nanostructured quantum magnet.” *Cell Rep. Phys. Sci.*, Mar. 2024. [\[Link\]](#)
8. Y. Xu, Z. Barani, P. Xiao, S. Sudhindra, **Y. Wang**, *et al.* “Crystal structure and thermoelectric properties of layered van der Waals semimetal ZrTiSe_4 .” *Chem. Mater.*, Sep. 2022. [\[Link\]](#)
9. **Y. Wang**, X. Chen. “Single crystal growth and electrochemical studies of garnet-type fast Li-ion conductors.” *Tungsten*, Sep. 2022. [\[Link\]](#)

MEDIA COVERAGE

- [Why LLZTO is the Secret to Super-Fast EV Charging](#) — UC Riverside, Feb. 2026
- [Cool Battery Power](#) — UC Riverside News, Oct. 2025
- [Solid electrolyte’s unique atomic structure helps next-generation batteries keep their cool](#) — Tech Xplore, Oct. 2025
- [ECE PhD Student Wins Materials Research Student Talk Award](#) — UCR ECE Department, May 2023